Dominance can move when ultimate-pair activity moves.

Centre for Blockchain Technologies

The Making of Dominant Vehicle Currencies

Evidence from DeFi

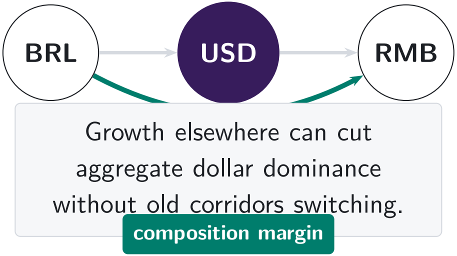
Jiahua Xu

Dominance changes when trade corridors change

TRADITIONAL-FINANCE INTUITION

A treasury desk keeps one thin dollar corridor while aggregate dollar share falls.

Growth elsewhere changes the denominator: route allocation, not only within-ultimate-pair substitution.



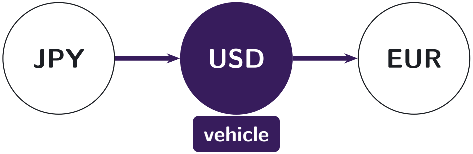
Note: The clearing example is an analogy for the denominator problem. The empirical object in this paper is the intermediary asset observed inside DEX pool routes.

Vehicle currencies are routes

FX ANALOGY

Yen–euro may trade through dollars.

Atomic-pair turnover data show two legs, not the customer route.



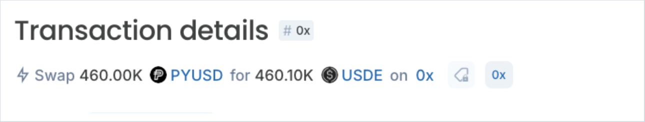
Pool logs reveal the path.

DEX data make the hidden route observable.

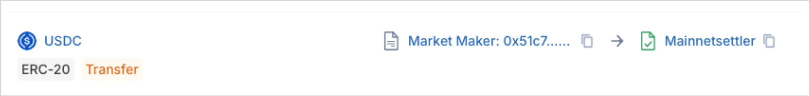
Note: Traditional FX records show turnover in each atomic currency pair, but not whether two legs belong to one customer route. The DEX setting preserves the transaction-level pool path, so the intermediary asset is observed directly.

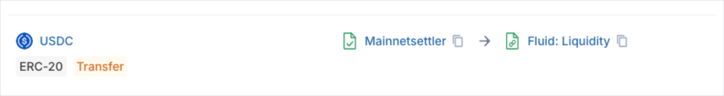
The PYUSD swap reaches a USDC-funded pool route

USER INSTRUCTION RECORDED BY THE EXPLORER



THE DECISIVE SETTLEMENT TRANSFERS



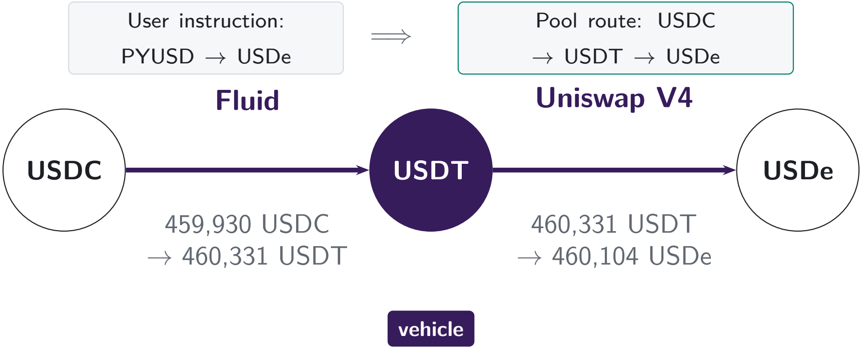


The instruction is PYUSD $\rightarrow$ USDe. The maker supplies the USDC used by the Fluid leg, which begins the connected pool route.

Note: The explorer screenshot and transfer trace refer to transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Inside the pools, USDT links USDC to USDe

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION



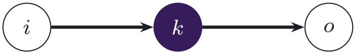
The observed ultimate trade is USDC → USDe. Its atomic trades are USDC → USDT and USDT → USDe; USDT is the vehicle that links them.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.

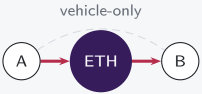


Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

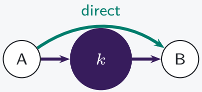
Architecture changes routing opportunities

V1 mandatory hub



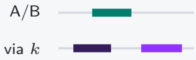
Hub use is imposed

V2 atomic-pair choice



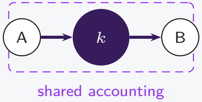
Hub use becomes a choice

V3 range choice



Atomic-pair opportunities shift

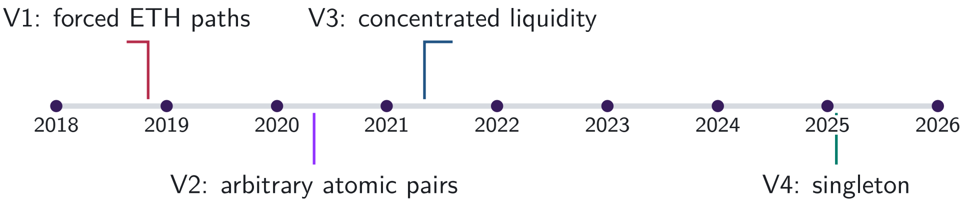
V4 net settlement



Settlement differs from route use

The route sample and architecture supplement answer different questions

DATA



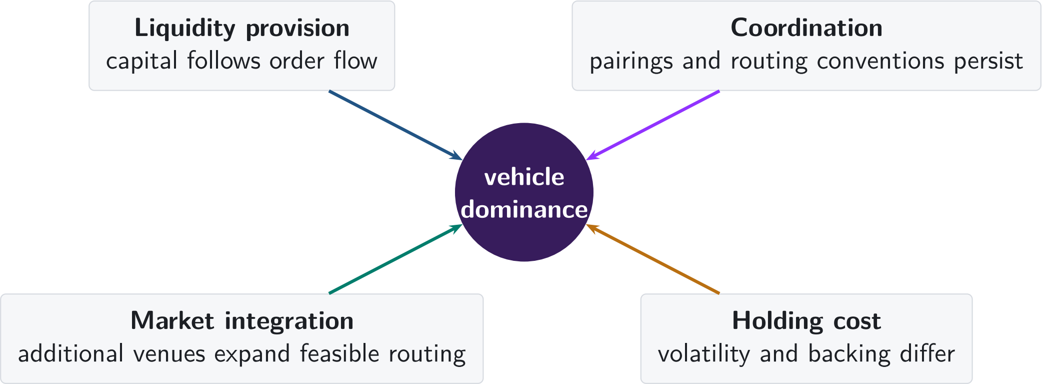
Main route sample

Atomic trades are linked within each transaction across the V2, V3, V4, Curve, and Balancer route perimeter.

V1/V2 architecture supplement

V1 forced-ETH paths and V2 atomic-pair formation isolate how protocol rules shape available routes.

Why might one asset become the dominant vehicle?



Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every pool is an ETH–token atomic pair, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: MARKET CHOICE

95.5%

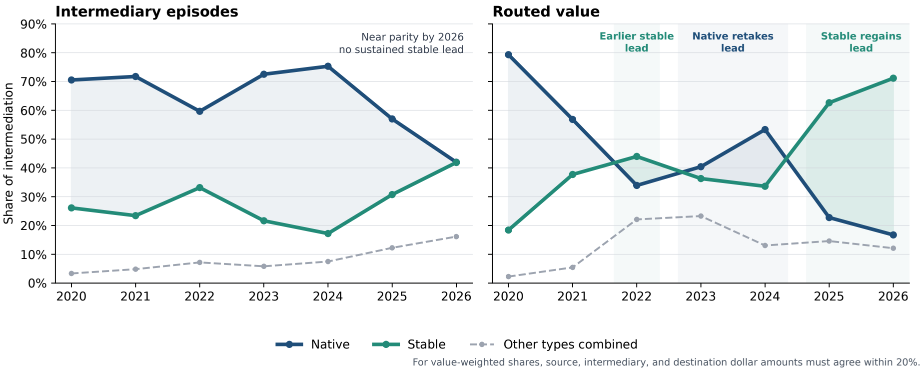
of single-leg V2 trades use a WETH pool in 2026

97.9%

of ultimate pairs first traded in 2026 include WETH

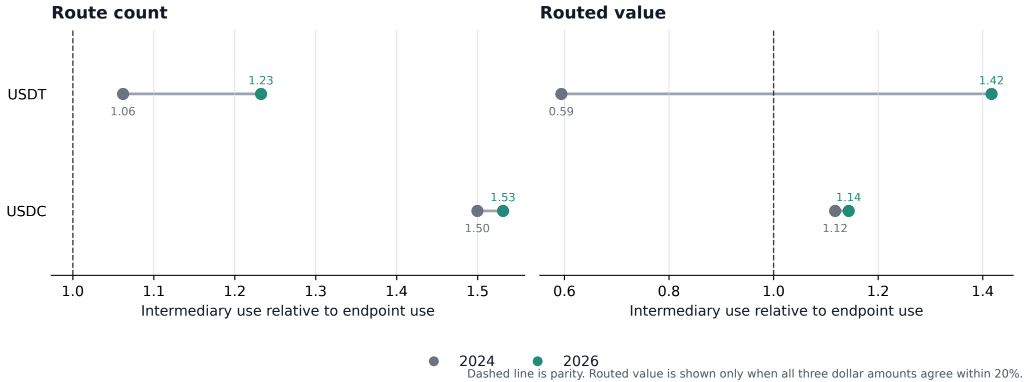
Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports single-leg trades and newly traded ultimate pairs on that venue.

Stablecoins regain the routed-value lead



Note: Exact two-atomic-trade routes. Count weights every route equally; value requires agreement among source, intermediary, and destination dollar values. The 2026 endpoint covers January through June.

USDT carries the stablecoin expansion



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

Note: Excess use divides intermediary share by route-endpoint share on the same perimeter. The figure compares the same January through June dates in 2024 and 2026.

Currency demand and vehicle use are compared within day

$$I_{a,t} = \alpha + \beta D_{a,t} + \mathbf{A}_a'\theta + \mu_t + \eta_a + \varepsilon_{a,t}$$

$I_{a,t}$	intermediary episode share	μ_t	date effects
$D_{a,t}$	own endpoint-demand share	η_a	currency effects
$\mathbf{A}_a$	asset-class indicators		SE clustered by date and currency

Note: Outcome and demand are shares of the same day's route universe, measured in percentage points (pp). The estimates are descriptive within-day associations.

Endpoint demand tracks vehicle use across currencies

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	$+34.56$	$+34.55$	$+0.05$	—
	(23.04)	(23.04)	(0.56)	—
Stablecoin	$+2.42^{*}$	$+2.42^{*}$	$+0.85^{*}$	—
	(1.32)	(1.32)	(0.50)	—
Endpoint-demand share ($D_{a,t}$)	—	—	$+1.59^{***}$	$+0.98^{***}$
	—	—	(0.03)	(0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

Endpoint-demand share passes through almost one for one after both fixed effects.

Note: 1,828,862 currency-days; 2,259 dates; currency/date-clustered standard errors. Class coefficients are percentage-point differences; the demand coefficient is percentage points of intermediary share per percentage point of endpoint-demand share. Asterisks $*$, $**$, and $***$ denote significance at 10%, 5%, and 1%, respectively.

WETH-linked ultimate pairs carry the count rotation

WETH

WETH is an endpoint of the ultimate pair

Stablecoins alone remain eligible as intermediaries.

All matched ultimate-pair groups

$+21.1$ **pp**

95% CI [$+18.2$ pp, $+24.0$ pp]

WETH-linked route-count mass: 26.6% $\rightarrow$ 46.8%

Ultimate-pair groups excluding WETH endpoints

$+3.7$ **pp**

95% CI [$+2.3$ pp, $+5.2$ pp]

79,347 matched groups

Within-ultimate-pair change: all groups $+0.2$ pp [-1.3 pp, $+1.7$ pp]; excluding WETH endpoints $+0.3$ pp [-1.9 pp, $+2.6$ pp].

Note: Exact two-atomic-trade native-WETH-versus-stablecoin routes on common month-days. The intervals use calendar HAC; within-ultimate-pair estimates fix ultimate pair, month-day, and route scope.

Trading shifts toward stablecoin-heavy ultimate pairs

Without WETH endpoints		Routed-value reallocation		Change within groups
$+21.5$ **pp**	$=$	$+23.5$ **pp**	$+$	-2.0 **pp**
95% CI [$+11.8$ pp, $+31.1$ pp]		95% CI [$+15.0$ pp, $+32.1$ pp]		95% CI [-9.8 pp, $+5.7$ pp]
strict-support stablecoin share of routed value Full matched sample: $+27.2$ pp [$+23.4$ pp, $+30.9$ pp]		Value shifts toward ultimate-pair and route-scope groups with higher stablecoin shares.		Stablecoin share changes within the same ultimate-pair and route-scope group.

Note: The daily midpoint identity is exact after removing WETH endpoints. Routed value requires agreement among source, intermediary, and destination dollar values.

Established ultimate pairs contribute little to the rotation

INTERMEDIARY CHOICE WITHIN CONTINUING ULTIMATE PAIRS

USDC → WBTC

largest positive ultimate-pair contribution

10.5% → **35.9%**

stablecoin share within that ultimate pair

-0.1 **pp**

aggregate within-ultimate-pair margin

Gross gains and losses across continuing ultimate pairs almost offset.

Note: Exact allocation of the pooled 2024-to-2026 route-count change on common calendar days. The named ultimate pair contributes $+0.05$ pp; all continuing ultimate pairs sum to -0.1 pp.

Growth favors ultimate pairs that already use stablecoins

ACTIVITY REALLOCATION ACROSS CONTINUING ULTIMATE PAIRS

USDT $\rightarrow$ WETH

largest positive ultimate-pair contribution

0.36% $\rightarrow$ **4.52%**

share of routed activity

$+8.6$ **pp**

aggregate reallocation margin

Routes on the named ultimate pair rise from 7,447 to 54,112.

Note: Exact allocation of the pooled route-count change. Ultimate-pair weights change across the same two endpoint periods while intermediary shares are held at their midpoint.

Newly traded ultimate pairs deliver the largest rotation margin

MARKETS APPEARING AFTER THE BASELINE PERIOD

USDS $\rightarrow$ WETH

largest named arrival

0 $\rightarrow$ 3,390

observed routes

$+21.0$ pp

aggregate new-ultimate-pair margin

Newly traded ultimate pairs account for most of the aggregate increase.

Note: Exact allocation of the pooled route-count change. Unlabelled assets contribute alongside the named ultimate pair; the full new-ultimate-pair population sums to $+21.0$ pp.

Local bridge depth predicts route choice

84.1%

route share captured by the deepest bridge

75.5% $\rightarrow$ **86.5%**

deepest-bridge share, 2024 to 2026

$+6.74$ **pp**

route-share slope after network-reach controls

The weaker atomic leg carries the economically relevant depth constraint.

Note: The analysis compares five potential vehicle assets within locally supported endpoint-date-route-scope sets. Depth is prior deposited capital on the weaker atomic leg; standard errors cluster by ordered ultimate pair and date.

A new stable bridge wins a foothold

WETH route only ➤ Stable bridge becomes feasible ➤ First 30 days of route flow

+7.93 pp
route-count share
SE 1.88 pp

+3.11 pp
supported-value share
SE 1.53 pp

92.6%
route counts retained by WETH

Availability produces partial reallocation while the incumbent remains dominant.

Note: 865 first-time events among ordered ultimate pairs previously routed only through WETH. Stable support requires positive prior-calendar deposited capital on both atomic pairs in Uniswap V2 or SushiSwap V2 and persistence over the next month. Share changes are paired against the prior 30 days; standard errors cluster by ordered ultimate pair and event date.

Higher capital predicts lower vehicle use at longer horizons

Deposited capital at t → Vehicle share at $t + \tau$

Next day	30 days	120 days
0.00 **pp**	-0.64 **pp**	-2.21 **pp**
SE 0.04 pp	SE 0.26 pp	SE 0.39 pp

Note: Daily panel of five vehicle assets over 2,238 V2 dates. Regressions absorb asset and date effects and use calendar-HAC inference. Coefficients are predictive associations.

Vehicle use barely changes subsequent capital

Vehicle share at t → Deposited capital at $t + \tau$

Next day

-0.03%

SE 0.03%

30 days

-0.32%

SE 0.27%

The smallest Holm-adjusted probability across the reciprocal short-horizon tests is 0.17.

Note: The same five-asset daily panel absorbs asset and date effects. Capital is V2 deposited reserves; provider flows and concentrated-liquidity depth are separate quantities.

New ultimate pairs made stablecoins dominant vehicles

1 **Intermediary choice persists within continuing ultimate pairs.**
Same ultimate pair, date, and route scope: $+0.2$ pp (SE 0.8 pp).

2 **Activity reallocates toward stablecoin-heavy ultimate pairs.**
Reallocation among continuing ultimate pairs contributes $+8.6$ pp to the count rotation.

3 **Newly traded ultimate pairs supply the largest margin.**
Ultimate-pair entry contributes $+21.0$ pp to the aggregate count rotation.

The dominant vehicle is made where ultimate pairs form and grow.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

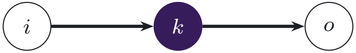
Robustness

- A7 Timing and venue-scope robustness
- A8 Count and value weighting
- A9 Observed route endpoints

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit

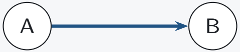


one route unit
two swap legs

- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

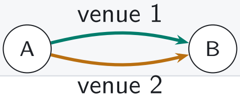
How we reconstruct a route from one transaction

Direct swap
one transaction

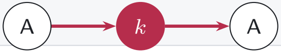


Sequential vehicle route
one transaction

Parallel split
one transaction

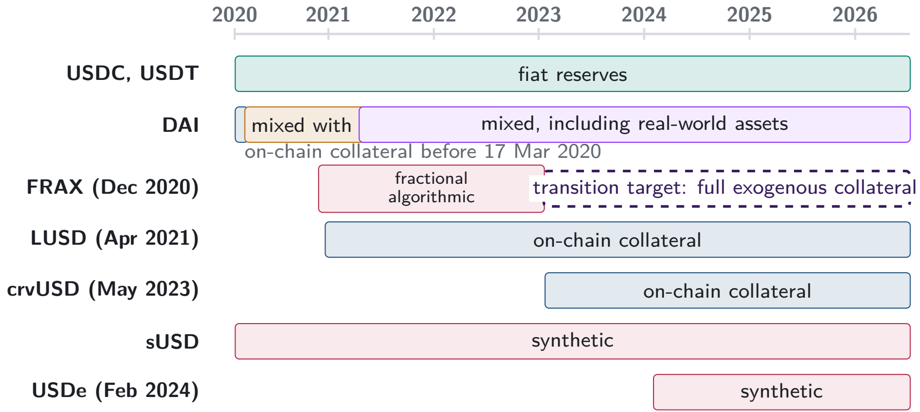


Round trip
one transaction



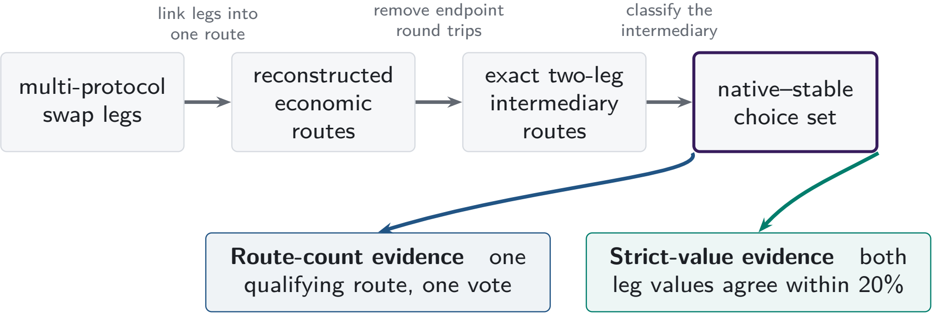
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths reveal that choice rather than leaving it inside aggregate turnover.

A2. Stablecoin backing changes over time



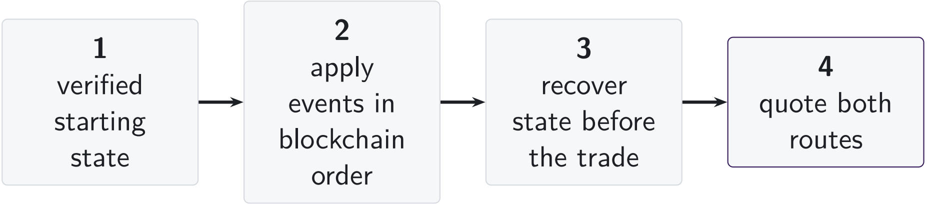
Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

A3. One route universe supports two measurements



Atomic pairs inside multi-asset pools can supply route legs. V1/V2 architecture evidence is separate; dollar support changes value weighting.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

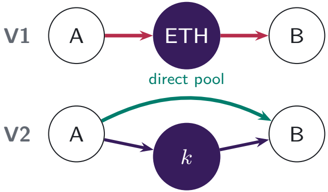
	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

The common object is the state immediately before execution. The fields required to recover that state vary with the pricing rule.

■ required state
□ hooks outside comparison

A5.1. Atomic-pair creation determines available paths

V1 PROTOCOL RULE ⟶ V2 MARKET CHOICE



V1 admits only ETH–token atomic pairs. V2 permits any ERC-20 atomic pair, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth comes from active liquidity

V3 CONCENTRATED-LIQUIDITY POSITIONS

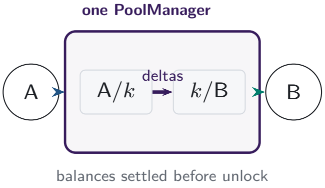
LPs choose a price range and fee tier for each atomic pool. Positions covering the current price determine active liquidity in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



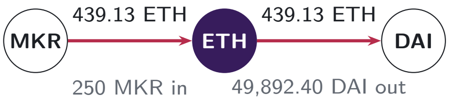
One PoolManager holds all pools. During a lock, the two atomic trades update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes the settlement boundary.

A6. V1 forced routes have an on-chain signature

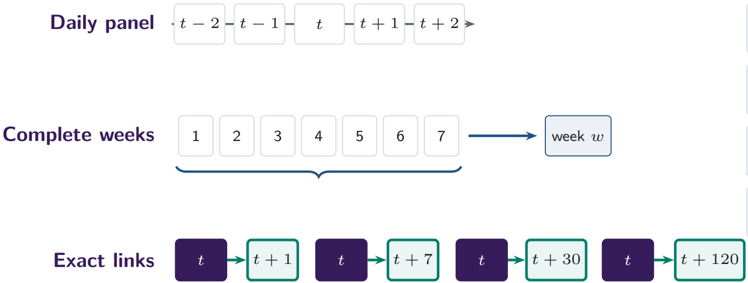
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction `0x4dca160d...96a0ca16` (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

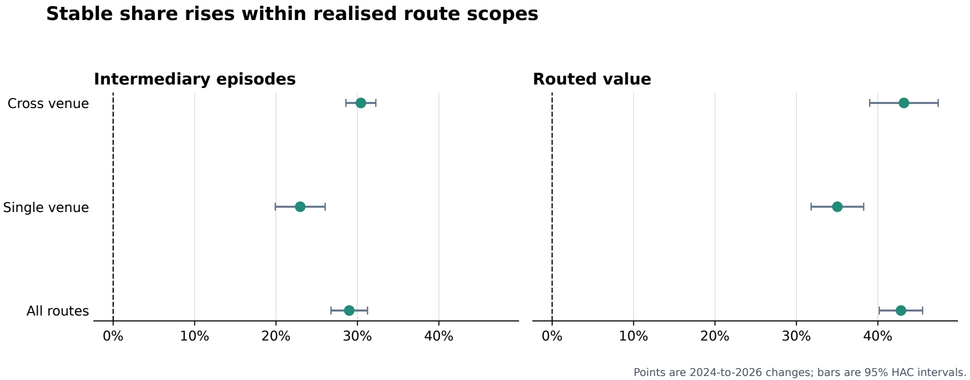
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

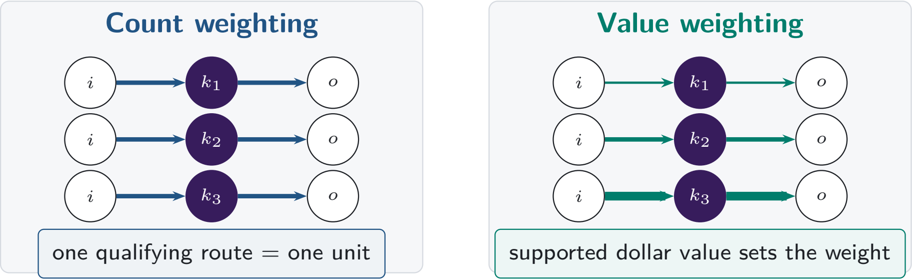
Exact horizons use the required target calendar date.

A7b. Stablecoin use rises within every venue scope



Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

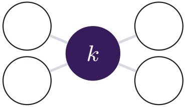
D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

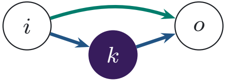
Excess vehicle use

$X_k = D_k \div E_k$
E_k: endpoint share of k

$X_k > 1$: use exceeds the endpoint benchmark



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I_{k,t}^{N} = \frac{n_{k,t}^{I}}{\sum_j n_{j,t}^{I}}$$

How often asset k sits inside a route.

Endpoint share

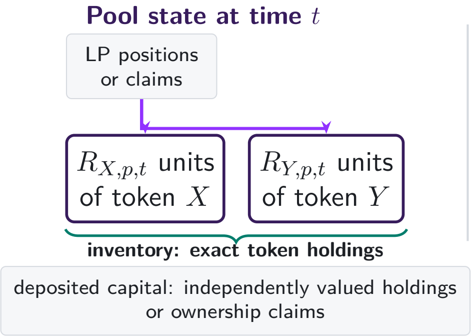
$$E_{k,t}^{N} = \frac{n_{k,t}^{E}}{\sum_j n_{j,t}^{E}}$$

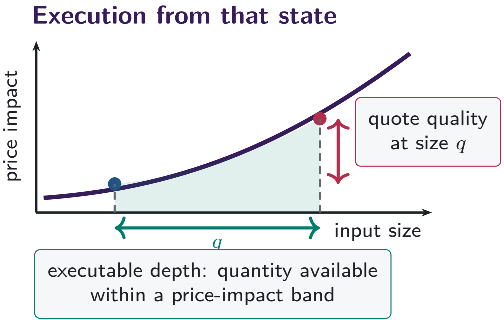
How often asset k appears at either end of the observed pool route.

Observed pool-route endpoints supply the benchmark.

Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ

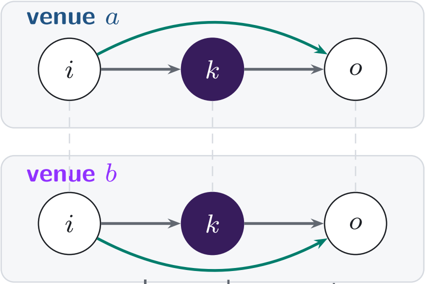




Depth and quotes vary with size and state.

A11. Additional venues expand the feasible route set

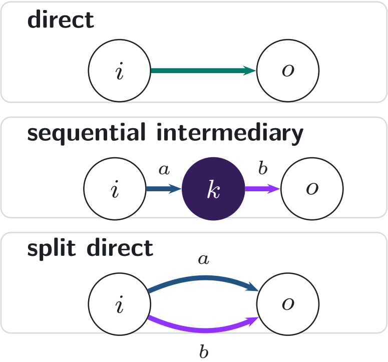
Feasible venue layers



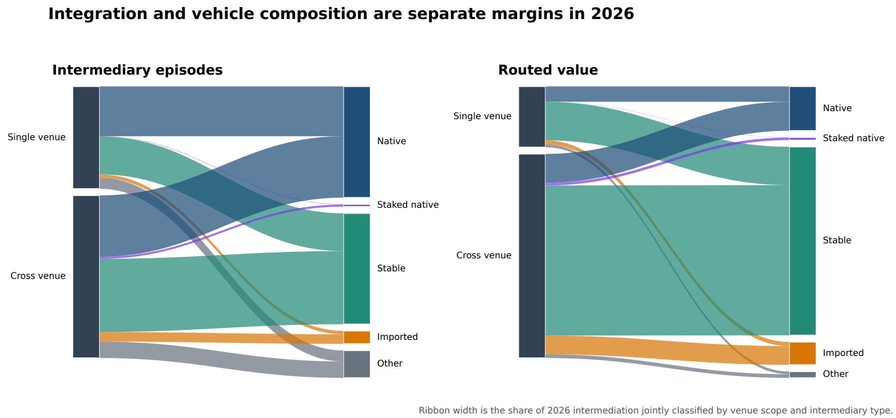
venue layers share assets;
available links define reach

realised
execution

Realised route topology



A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage